SIMATS ENGINEERING (SAVEETHA SCHOOL OF ENGINEERING)

Department of Artificial Intelligence and Data Science

ASSESSMENT TOOL 2 – SCENARIO BASED

Course Code:	DSA0305	**Course Title:**	Natural Language Processing
CO Assessed:	CO2 – Manipulation	**Assessment:**	Assessment Tool 2 – SCENARIO BASED
Weightage:	30% of CIA		
CO5 Statement:	Design inflectional, derivational, finite-state parsing, stemming algorithms and for analyse morphological methods. (BL-3)		
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Section / Batch:	CSE-AI	**Date:**	31-07-2026

Code of Conduct

*I **P.KALYANI (192472199)**(Name / Reg No) certify that this submission is my original work and that I have adhered to the guidelines specified for this assessment. I understand that any violation of academic integrity rules will result in disciplinary action.*

Signature of the Student: **P.Kalyani**

Instructions

- Read all questions carefully before attempting the answers.
- Answer all five questions. Each question carries equal marks.
- Show all intermediate steps, calculations, probability computations, tagging decisions, and justifications wherever applicable.
- Duration: 30 minutes.

Section / Topic	Focus	Q Nos
English Word Classes, Tagsets for English, Part of Speech Tagging	Morphological decomposition of analyzing, analysis, and analytical; identification of roots, affixes, inflectional and derivational forms, and normalization for indexing.	Q1
English Word Classes, Tagsets for English, Stochastic Part of Speech Tagging	Morphological parsing of disagree, agreement, and agreeable; identification of prefixes, suffixes, base forms, and semantic effects of derivation.	Q2

Counting Words, English Word Classes, Part of Speech Tagging	Morphology-based normalization of govern, government, and governance for topic modeling, clustering, and lexical standardization.	Q3
Rule-Based Part of Speech Tagging, Transformation-Based Tagging	Morphological analysis of activate, activation, and reactivation; identification of derivational layers, word-class changes, and semantic shifts.	Q4
Rule-Based Part of Speech Tagging, English Word Classes, Part of Speech Tagging	Inflectional normalization of create, creates, and creating; extraction of root forms and classification of tense/aspect transformations.	Q5

Annexure A – SCENARIO BASED

1. A large-scale search engine indexing system processes user queries containing word variants such as “analyzing”, “analysis”, and “analytical”. The system must improve retrieval accuracy by grouping semantically related terms under a unified representation. Design a morphological processing approach to decompose each input word into root and affixes, distinguish between inflectional and derivational forms, and normalize them for indexing. Apply the process to the given inputs and determine the root form along with the morphological structure of each word.

2. An AI-based sentiment analysis platform receives product reviews containing words like “disagree”, “agreement”, and “agreeable”. The system must interpret sentiment consistently despite variations in word formation. Develop a morphological parsing strategy to identify prefixes, suffixes, and base forms, while capturing how derivational changes influence meaning. Process the given inputs to extract roots and classify each transformation, ensuring accurate semantic interpretation across all word forms.

3. A text analytics pipeline in a news aggregation system handles documents containing terms such as “govern”, “government”, and “governance”. The system must standardize these variations for topic modeling and clustering. Construct a morphology-based normalization mechanism that decomposes each word into its root and affixes, identifies derivational layers, and maps them to a common base form. Apply the approach to the input words and derive the normalized representation along with their morphological breakdown.

4.A document classification system processes input words such as “activate”, “activation”, and “reactivation” to improve semantic grouping and indexing.
Construct a morphological parsing strategy to identify prefixes, suffixes, and root forms while distinguishing derivational layers.
Analyze how prefix addition and suffix transformation affect meaning and word class.
Apply the process to decompose each word and map them to a unified base representation. Derive the morphological structure and normalized root for each input word.

5. A search optimization module processes input words such as “create”, “creates”, and “creating” to ensure consistent retrieval across grammatical variations. Design a morphology-based normalization approach focusing on inflectional transformations such as tense and aspect.
Identify suffix patterns and map all variations to a common base form without altering meaning.
Apply the process to extract root forms and classify each transformation. Determine the morphological breakdown and normalized output for each word.

Rubrics for Calculation

Criteria	Weightage (%)	Level 4 (Exemplary)	Level 3 (Proficient)	Level 2 (Developing)	Level 1 (Beginning)
Understanding of Scenario	20%	Demonstrates clear and complete understanding of the scenario and context	Shows good understanding with minor gaps	Partial understanding; misses key aspects	Misinterprets or fails to understand the scenario
Identification of Key Issues	20%	Accurately identifies all relevant issues and factors involved	Identifies most key issues with minor omissions	Identifies limited issues; some irrelevant points	Unable to identify key issues
Application of Concepts	25%	Applies appropriate concepts effectively to address the scenario	Applies concepts with minor errors	Limited or partially correct application	Unable to apply relevant concepts
Decision Making	20%	Makes well-reasoned, logical decisions with strong rationale	Makes reasonable decisions with some justification	Makes weak decisions with limited justification	No clear decisions made or rationale provided
Clarity of Response	15%	Response is clear, structured, and logically presented	Generally clear with minor issues	Somewhat unclear or poorly structured	Disorganized and difficult to understand

Q. No. / Criteria Level	Understanding of Scenario	Identification of Key Issues	Application of Concepts	Decision Making	Clarity of Response	Sub. Total	Total %
1							
2							
3							
4							
5							

Faculty In-charge	Course Coordinator	Program Director
Signature: _______________	Signature: _______________	Signature: _______________
Date: ___________________	Date: ___________________	Date: ___________________

1. Search Engine Indexing System

Scenario Understanding

A large-scale search engine stores millions of documents. Users often search using different forms of the same word, such as analyzing, analysis, and analytical. If these words are indexed separately, relevant documents may not be retrieved, reducing search accuracy. Morphological processing helps identify the common root word so that all related terms are grouped together. This improves indexing efficiency, search relevance, and overall retrieval performance.

Key Issues Identified

- Presence of multiple word variants representing the same concept.
- Need to identify the root word accurately.
- Distinguish inflectional changes (grammar only) from derivational changes (new word formation).
- Normalize all variants before document indexing.
- Maintain semantic consistency across the search engine.

Morphological Processing Approach

1. Receive the input word.
2. Perform tokenization.
3. Identify prefixes and suffixes.
4. Separate the root from affixes.
5. Classify the transformation as inflectional or derivational.
6. Convert every variation into its normalized base form.
7. Store only the normalized root in the search index.

Morphological Analysis

Input Word	Root	Prefix	Suffix	Type	Word Class	Normalized Form
analyzing	analyze	—	ing	Inflectional	Verb	analyze
analysis	analyze	—	sis	Derivational	Noun	analyze
analytical	analyze	—	tic + al	Derivational	Adjective	analyze

Decision

The search engine indexes all three words under the single base word analyze. During user searches, any variation automatically maps to this root, ensuring that all relevant documents are retrieved regardless of the word form used.

Advantages

- Improves search accuracy.
- Reduces duplicate index entries.
- Enhances document retrieval speed.
- Increases semantic consistency.
- Supports efficient information retrieval.

Conclusion

Morphological normalization successfully groups all related forms under analyze, allowing the search engine to return more complete and relevant search results.

2. AI Sentiment Analysis Platform

Scenario Understanding

Sentiment analysis systems must correctly interpret reviews even when users use different forms of the same word. For example, disagree, agreement, and agreeable all originate from the root agree, but prefixes and suffixes modify their meaning or grammatical category. Morphological parsing enables the system to understand these relationships and improve sentiment classification.

Key Issues Identified

- Detect prefixes and suffixes.
- Extract the base word.
- Identify derivational transformations.
- Preserve semantic meaning during normalization.
- Improve sentiment prediction consistency.

Morphological Parsing Strategy

1. Read each input word.
2. Detect prefixes.
3. Detect suffixes.
4. Extract the base form.
5. Determine the effect of the transformation.
6. Store the normalized root while preserving semantic information.

Morphological Analysis

Input Word	Prefix	Root	Suffix	Transformation	Word Class	Meaning
disagree	dis	agree	—	Derivational	Verb	Opposite action
agreement	—	agree	ment	Derivational	Noun	State of agreeing
agreeable	—	agree	able	Derivational	Adjective	Pleasant or acceptable

Semantic Interpretation

- **dis-** reverses the meaning.
- **-ment** converts the verb into a noun.
- **-able** converts the verb into an adjective.

Decision

The system stores agree as the normalized root while recording derivational information separately for sentiment interpretation.

Benefits

- Improves sentiment accuracy.
- Reduces vocabulary size.
- Maintains semantic consistency.
- Handles unseen word variants effectively.

Conclusion

Morphological parsing enables the AI system to understand related word forms and correctly identify sentiment despite changes in word formation.

3. News Aggregation System

Scenario Understanding

News articles frequently contain different forms of the word govern, such as government and governance. If these words are treated independently, documents discussing the same topic may not be grouped together. Morphological normalization solves this issue by mapping all related words to a common root.

Key Issues

- Topic clustering requires normalized vocabulary.
- Derivational suffixes create different word classes.
- Common representation improves document similarity.
- Root extraction increases clustering accuracy.

Normalization Mechanism

1. Tokenize the input.
2. Separate root and suffix.
3. Identify derivational layers.
4. Normalize to the common root.
5. Store normalized representation.

Morphological Breakdown

Word	Root	Suffix	Type	Word Class	Normalized Form
govern	govern	—	Root	Verb	govern
government	govern	ment	Derivational	Noun	govern
governance	govern	ance	Derivational	Noun	govern

Decision

All document keywords are indexed under govern, ensuring that documents discussing governance or government are grouped into the same topic.

Advantages

- Better topic modeling.
- Improved document clustering.
- Reduced vocabulary complexity.
- Faster information retrieval.
- Better semantic matching.

Conclusion

The normalization mechanism ensures that all related words contribute to the same topic representation, improving clustering and document organization.

4. Document Classification System

Scenario Understanding

Document classification requires identifying words with similar meanings even when prefixes and suffixes modify them. Words like activate, activation, and reactivation belong to the same semantic family but differ in grammatical structure. Morphological parsing groups them together.

Key Issues

- Detect derivational prefixes.
- Identify suffix transformations.
- Extract the root word.
- Normalize derived words.
- Preserve semantic relationships.

Morphological Parsing Strategy

1. Identify prefix.
2. Detect suffix.
3. Extract root.
4. Determine derivational layers.
5. Normalize to the base word.

Morphological Analysis

Word	Prefix	Root	Suffix	Type	Word Class	Normalized Form
activate	—	activate	—	Base	Verb	activate
activation	—	activate	ion	Derivational	Noun	activate
reactivation	re	activate	ion	Derivational	Noun	activate

Meaning Analysis

- **activate** → perform an action.
- **activation** → the process of activating.
- **reactivation** → activating again.

Decision

The classifier groups every variation under activate while retaining derivational information for classification.

Benefits

- Better semantic grouping.
- Higher document classification accuracy.
- Reduced feature dimensionality.
- Improved indexing efficiency.

Conclusion

Morphological normalization allows different derived forms to be treated as one concept, improving document classification performance.

5. Search Optimization Module

Scenario Understanding

Search engines often receive grammatical variations of the same word. Words such as create, creates, and creating differ only in tense or aspect. Inflectional morphology removes these grammatical endings while preserving meaning.

Key Issues

- Detect inflectional suffixes.
- Preserve the original meaning.
- Normalize every grammatical variation.
- Improve retrieval consistency.

Normalization Approach

1. Identify inflectional suffixes.
2. Remove the suffix.
3. Recover the base form.
4. Store only the normalized word.

Morphological Analysis

Word	Root	Suffix	Transformation	Word Class	Normalized Form
create	create	—	Base	Verb	create
creates	create	s	Present tense	Verb	create
creating	create	ing	Progressive aspect	Verb	create

Decision

All grammatical variations are indexed using the single root **create**, ensuring that users receive the same search results regardless of the word form entered.

Advantages

- Improves retrieval accuracy.
- Eliminates duplicate indexing.
- Supports stemming and lemmatization.
- Enhances search efficiency.
- Maintains grammatical consistency.

Conclusion

Inflectional normalization maps all grammatical forms to the common base word create, resulting in more accurate and consistent search results.